

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W005
Well Name	SYN-Assam-Duliajan-005
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.355745 N
Longitude	95.261269 E
Well Status	Completed
Well Role	Historical (offset reference well)
Spud Date	28-Nov-2020
Drilling Completed	09-Dec-2020
Report / Completion Date	16-Dec-2020
Total Depth (Driller)	3,560 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-11
Drilling Days	12 days
Recorded Events	5
Recorded NPT	91.9 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W005
Well Name: SYN-Assam-Duliajan-005
Field: SYN-Assam-Duliajan
Latitude: 27.355745
Longitude: 95.261269
Total Depth: 3,560 m
Primary Formation: Sandstone
Well Status: Completed
Spud Date: 2020-11-28
Event Count: 5
Total NPT: 91.9 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W005 (SYN-Assam-Duliajan-005) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,560 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,560 m, completed in 12 rotary drilling days between 28-Nov-2020 and 09-Dec-2020. Average overall penetration rate for the well was 12.4 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–782 m), Shale (782–2,008 m), Sandstone (2,008–3,176 m), Limestone (3,176–3,560 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 5 reportable events were recorded, comprising 2 × Wellbore Instability, 2 × Mud Loss, 1 × Cementing Issue. Aggregate non-productive time was 91.9 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 780 m, 9-5/8" at 2,010 m, 7" at 3,560 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Completed**. The well was completed on the primary sandstone objective and handed over for production testing.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W002	SYN-Assam-Duliajan-002	4.61	3,450	Sandstone	7
W010	SYN-Assam-Duliajan-010	4.86	3,260	Sandstone	7
W008	SYN-Assam-Duliajan-008	5.89	3,690	Sandstone	8
W001	SYN-Assam-Duliajan-001	7.18	3,470	Sandstone	4
W003	SYN-Assam-Duliajan-003	7.45	3,850	Sandstone	8
W007	SYN-Assam-Duliajan-007	9.05	3,400	Sandstone	3

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	782	782	Routine drilling, no reportable event
2	Shale	782	2,008	1,226	Wellbore Instability recorded (1 event)
3	Sandstone	2,008	3,176	1,168	Mud Loss, Wellbore Instability, Cementing Issue recorded (4 events)
4	Limestone	3,176	3,560	384	Routine drilling, no reportable event

Formation: Claystone Top: 0 m Bottom: 782 m
Formation: Shale Top: 782 m Bottom: 2008 m
Formation: Sandstone Top: 2008 m Bottom: 3176 m
Formation: Limestone Top: 3176 m Bottom: 3560 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–782	9.58	9.1–10.1	31.7	140	8.1
Shale	782–2,008	11.10	10.2–12.0	16.3	125	17.6
Sandstone	2,008–3,176	12.58	11.7–13.5	15.3	111	25.5
Limestone	3,176–3,560	14.04	13.8–14.2	7.6	91	35.7

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 14.0 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 120 rpm range on average with weight on bit averaging 20.1 klbf. Penetration rate declined with depth from 31.7 m/hr in the upper hole to 7.6 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0036	916	Shale	Wellbore Instability	High	23.8
E0037	2,159	Sandstone	Mud Loss	High	11.1
E0038	2,191	Sandstone	Wellbore Instability	Medium	11.6
E0039	2,278	Sandstone	Cementing Issue	Critical	42.5
E0035	2,296	Sandstone	Mud Loss	Low	2.9

Well ID: W005		Event Type: Wellbore Instability		Depth: 916 m		Formation: Shale		Severity: High		Duration: 23.8 hours
Well ID: W005		Event Type: Mud Loss		Depth: 2159 m		Formation: Sandstone		Severity: High		Duration: 11.1 hours
Well ID: W005		Event Type: Wellbore Instability		Depth: 2191 m		Formation: Sandstone		Severity: Medium		Duration: 11.6 hours
Well ID: W005		Event Type: Cementing Issue		Depth: 2278 m		Formation: Sandstone		Severity: Critical		Duration: 42.5 hours
Well ID: W005		Event Type: Mud Loss		Depth: 2296 m		Formation: Sandstone		Severity: Low		Duration: 2.9 hours

7.1 EVENT DETAIL (1–3 OF 5)

EVENT E0036 — WELLBORE INSTABILITY AT 916 m (SHALE)

Well ID: W005
Event ID: E0036
Event Type: Wellbore Instability
Depth: 916 m
Formation: Shale
Severity: High
Duration: 23.8 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Sloughing Shale at 916 m produced excess cuttings volume and hole enlargement; mud properties adjusted. Parameters recorded around the event depth: mud weight 10.43 ppg, ROP 17.7 m/hr, 132 rpm, WOB 15.1 klbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

EVENT E0037 — MUD LOSS AT 2,159 m (SANDSTONE)

Well ID: W005
Event ID: E0037
Event Type: Mud Loss
Depth: 2159 m
Formation: Sandstone
Severity: High
Duration: 11.1 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Seepage-to-partial mud losses observed while drilling Sandstone at 2159 m; mud weight trimmed and losses monitored. Parameters recorded around the event depth: mud weight 11.77 ppg, ROP 17.9 m/hr, 112 rpm, WOB 21.1 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0038 — WELLBORE INSTABILITY AT 2,191 m (SANDSTONE)

Well ID: W005
Event ID: E0038
Event Type: Wellbore Instability
Depth: 2191 m
Formation: Sandstone
Severity: Medium
Duration: 11.6 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Sloughing Sandstone at 2191 m produced excess cuttings volume and hole enlargement; mud properties adjusted. Parameters recorded around the event depth: mud weight 11.92 ppg, ROP 16.4 m/hr, 116 rpm, WOB 22.0 klbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

7.2 EVENT DETAIL (4–5 OF 5)

EVENT E0039 — CEMENTING ISSUE AT 2,278 m (SANDSTONE)
Well ID: W005 Event ID: E0039 Event Type: Cementing Issue Depth: 2278 m Formation: Sandstone Severity: Critical Duration: 42.5 hours Operational Impact: Major non-productive time; section objectives affected and specialist intervention required. Description. Incomplete annular cement coverage near 2278 m in Sandstone; remedial cementing operation carried out. Parameters recorded around the event depth: mud weight 12.15 ppg, ROP 16.7 m/hr, 114 rpm, WOB 23.0 klb. Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage. Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage. Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.
EVENT E0035 — MUD LOSS AT 2,296 m (SANDSTONE)
Well ID: W005 Event ID: E0035 Event Type: Mud Loss Depth: 2296 m Formation: Sandstone Severity: Low Duration: 2.9 hours Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment. Description. Loss of circulation into porous Sandstone at 2296 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 12.16 ppg, ROP 17.1 m/hr, 113 rpm, WOB 23.4 klb. Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval. Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	780	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	2,010	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 660 m
7"	N-80, 29 lb/ft	3,560	8-1/2"	Tail 15.8 ppg, TOC 1,860 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0039** — Cementing Issue at 2,278 m in Sandstone (Critical, 42.5 hr). Incomplete annular cement coverage near 2278 m in Sandstone; remedial cementing operation carried out.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,560 m
Primary Formation	Sandstone
Major Events	3 of 5 events rated High or Critical
Total NPT	91.9 hours
Completion Status	Completed
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Wellbore Instability** at 916 m, 2,191 m (Shale, Sandstone) — 35.4 hr cumulative.
- **Mud Loss** at 2,159 m, 2,296 m (Sandstone) — 14.0 hr cumulative.
- **Cementing Issue** at 2,278 m (Sandstone) — 42.5 hr cumulative.

9.2 Key Historical Observations

- Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Raise inhibition before entering the Shale/Sandstone section and limit open-hole exposure time; treat rising cavings volume as an early warning.
- Approach the Sandstone interval (2,159–2,296 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.
- Review slurry design, centralisation and displacement rate for the stage covering 2,278 m; run a bond log before proceeding.

10. DOCUMENT CONTROL

Document Number	WCR-W005-R0
Revision	R0
Issue Date	16-Dec-2020
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_ formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.